Design for Society



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Abstract This chapter presents the "Design for Society" concept and provides a backdrop for the papers included as chapters in this section of the book. We suggest that the Design for Society concept can be viewed as an umbrella for a range of concepts related to how we can leverage the principles of design research and practice to create holistic solutions to societal challenges. Design can drive societal innovation, and the collection of papers we have compiled below focus on how design can enable society to become more equitable and inclusive. Our hope is that this chapter and the subsequent chapters might inspire design practitioners and researchers to learn from and build on each other's ideas - as opposed to reinventing the wheel or even making the same mistake, so that we may move forward in tackling our societal challenges and achieving the UN SDGs.

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Since the turn of the century, new societal challenges have continued to arise, making it all the more clear that social, economic, and environmental issues cannot be dealt with separately, as was described in the Brundtland Report published by the UN World Commission on Environment and Development (also known as Our Common Future) [1]. In this section, we present a series of papers that highlight the emerging role that Design for Society has in enabling a holistic approach to tackling these challenges.

Today we live in a "Designed World", a human-made artificial world in which design has been involved in every aspect of creating our living environments [2]. However, design as a creative act is intertwined with the act of destruction as everything created requires something else to be changed [2]. Some argue that the societal challenges we are facing today can be seen as a result of "user-centred" design in which the focus was merely on fulfilling short-term human-centred desires without taking into consideration the long-term destructive effects on the environment and society [2]. Nigel Whiteley in his book, *Design for Society* (1993), pointed out the problems of consumer-oriented design in which the design process generally did not go beyond fulfilling the immediate client's requirements, thereby ignoring the greater environmental impact, which in the long run led to environmental unsustainability. Looking forward, the question arises as to how we instead can leverage the principles of design research and practice to create solutions to not only help "fix" the problem but also to regenerate the planet and society through "destroying" harmful behaviours and practices.

Already today there are signals that we are moving in this direction where the term "Design for Society" takes on a broader meaning and could serve as an umbrella for a range of concepts. For example, design for inclusion strives to create equitable products and services by embracing the opportunities and challenges presented by a diverse population. Human-centered thinking plays a central role in developing and delivering solutions that are not only accessible but that also provide enriching and engaging experiences for as many people as possible, regardless of age, gender and (dis)ability. Further, concepts such as design for social innovation and design for sustainability walk hand in hand. Design for sustainability tends to have a more technical approach to the intervention and construction of the artificial world while design for social innovation tends to have a more "liquid" and enabling intervention in the way people organize and are called to act on behalf of a common well-being. Several strengths of design for social innovation projects, such as understanding user experiences, rapid prototyping, visualisation and systematic approach, have been illustrated by Mulgan [3] [4]. Within design for sustainability, these same tools can be used to design regenerate and repair actions to occur in tandem. Today numerous designers are working to diminish the ecological footprint of products and services throughout their lifecycle by designing for recycle, reuse and repair instead of producing new "future" waste while also striving to reduce consumption excesses

through designing new circular and sharing economy models. Other areas include supporting and designing suitable processes to assist self-organized citizens in their own efforts to save and regenerate our common home.

Yet it is important to remember that design has several levels. For example, the Danish Design Ladder or the Design Management staircase [5, 6] describes design as having three levels: (1) at an operational level, design is for achieving a specific task, (2) at the functional level, design is applied as a process or discipline; and (3) at a strategic level, design is holistically used for strategic intent [4]. We view Design for Society as incorporating all three levels with a particular focus on the strategic level in which a systems approach should be used to analyze potential interactions among actors, and even with existing and emerging technologies, and their effects on the environment and society. This analysis should include not only the "here-and-now", but it should also be intertwined with scenario thinking, in which an analysis of how potential possible futures might play out is conducted to better understand the ramifications of one's design efforts today.

The above is a challenging task, and as has been noted by several authors, design, and in particular design for social innovation, is wrought with complications, such as the use of expensive methods and tools, lack of awareness of conditions or limitations beyond the project, and lack of resources, especially by entrepreneurs designing new solutions [4]. Another considerable challenge is the difficulty to learn from others' experiences working with design for society projects. It is with this challenge in mind that we have assembled this series of papers. Our hope is that these papers might inspire design practitioners and researchers to learn from and build on each other's ideas—as opposed to reinventing the wheel or even making the same mistake, thereby wasting valuable resources. In this manner, we hope to move forward in fulfilling the UN's Agenda 2030 goals. Thus, the collection of papers we have assembled here as chapters includes a number of topics related to design research for societal issues, the environment, and sustainability, and they present us with challenges, ideas and some solutions. Design can drive societal innovation, and this collection of papers demonstrates just that. Whether addressing globalization issues, implementing social change, finding alternative means of transportation, bringing innovation to traditional organizations, making use of global sustainable practices and circular business models, allowing access to play for all or facing the challenges of smart mobility, design is the key. These papers focus on redesigning society and making it more equitable and inclusive, and they give us a new perspective on what design can bring to society in terms of new ways of living.

Appendix: Chapters

Addressing Glocalization Challenges Through Design-Driven Innovation Approaches

The *Addressing Glocalization Challenges through Design-driven Innovation Approaches* suggests a framework for design-driven glocalization, making use of seven case studies from all over the world, cross comparing them to other design-driven approaches. Globalization has brought the opportunity for companies to be present in international markets. However, Globalization strategies, adaptation strategies and even Localization strategies although dynamic, have created continuous tensions and are of considerable consequence for corporate strategy decision making and processes.

The option for a Global strategy or Glocalization has its challenge as well but it is of relevance to the UN 2030 sustainable development goals. Design driven glocalization can lead to the development of sustainable cities and communities, it can preserve local cultural heritage, maintain cultural atmosphere of the local economies and even reduce environmental impact. It can also build work opportunities and economic growth.

It can also result in the support of local craftsman from rural areas, it can help build resilient infrastructure and innovation. Allowing for the investment in new technologies that focus on sustainability, responsible consumption and production. It can also allow for the rethinking of the fast street fashion system that is highly based on a product's fast lifecycle and planned obsolescence.

Inclusive Design as Promoter of Social Transformations: Understanding Androgyny in Contemporary Society

This chapter contributes to deepen the knowledge about the diversity of which contemporary society is constituted. This is a challenge for the design to constantly renew itself and meet people's expectations. Minorities and people who stray from some common 'norm' or stereotype are often the target of prejudice and discrimination—problems that Design can help to alleviate. In this article some questions about gender identity and proposals about genderless / androgyny are addressed.

This paper addresses the role of Design in the fight against stereotypes and prejudices related to gender issues and the way it influences design knowledge and consumer choices. This can directly contribute to SDG No. 5 'Achieve gender equality and empower all women and girls' and, in general, to SDG No. 16—promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels'.

Dynamic Ride Sharing As An Alternative Transportation Mode For Commuting Among Metu Campus And Eryaman

The paper presents us with a study of TAG app (let's ride in a single car) and is very interesting for Design for society. The work clearly explained, and the methods are very well described. The paper will allow other research to structure a similar research because the process is all documented and detailed. The authors were able to present the findings in a very consistently way.

Dynamic ride sharing is an alternative way of commuting to individual car use with the utilization of improving available seat capacity for new riders through which more sustainable trips are ensured among drivers and riders who have similar itineraries in closer time slots. The main objective of the study was to acquire in depth knowledge about the phases of dynamic ride sharing practices, experiences of individuals commuting via ride sharing, their concerns and acquisitions from ride sharing, and to determine potential solutions for improving the overall ride sharing practice which would make it an effective and preferable way of alternative transportation in a sustainable manner. The study revealed user's underlying motivations towards ride sharing, their concerns, needs and desires which would help generate better approaches to performing ride sharing practice, in terms of both optimized and secured ways for better experiences in the light of fresh insights gathered in specific themes.

This chapter highlights a lot of challenges for society. Mode choice decisions for travelling depend on many different issues such as convenience, comfort, safety, time considerations, accessibility and cost, yet these decisions tend to be affected by social, economic and environmental impacts of commuters' acts and behaviours. It seems that there has been substantial interest in promoting sustainable transportation alternatives due to climate change, aging infrastructure, and raising green culture. Population-based changes in individuals' behaviours, attitudes and knowledge are important to accomplish wide-spread adoption of alternative transportation modes. Moreover, for the improvement in the quality of life of campus residents, some measurements should be implemented through sustainable transportation which will return in many social, economic and environmental benefits. In this respect, dynamic ride sharing could be one of the most prominent solutions for METU Campus where hitchhiking culture is truly favoured among all its members.

There are several economic, environmental and social benefits of dynamic ride sharing. In general terms and considering identifiable objectives for the sustainable development for UN, as current transportation models are not both environmentally and socially sustainable, road transportation is one of the most challenging issues of the new world. In one respect, car pollution directly causes global climate change as being one of the major sources of greenhouse emissions. In another respect, traffic congestions have a great influence on the reduction of people's quality of life worldwide. In this respect, many researchers from diverse disciplines such as transportation, economics, and behavioural, social and environmental psychology have described dynamic ride sharing as an effective solution to the inefficiency of

current transportation models. Moreover, circumstances such as finite oil reserves, rising gas prices, traffic congestions and related environmental concerns have given a direction to people to use their personal automobiles more wisely with an increased interest in services such as dynamic ride sharing. Further-more, ride sharing allows participants having automobiles to share travel expenses while offering those without cars an enhanced mobility. In overall ride sharing, in other terms the joint travel of two or more persons in a single car has initiated a common way of sharing also the costs and benefits together of a shared private car.

Castelo Branco Embroidery—Tradition and Innovation

The chapter *Castelo Branco Embroidery—Tradition and Innovation* touches on both sustainability of materials and of tradition. It deals with the cultural heritage of Castelo Branco, a region of the Portuguese country as well as it touches on the area of sustainability.

This study presents a process of developing garments with the application of Castelo Branco Embroidery. It describes three different phases: bibliographical revision Embroidery of Castelo Branco, the brief that was present within a Design Challenge to fashion designers, and the products resultant from the proposed brief. The authors describe deeply the materials and types of stiches, colours, motifs and symbology of the Castelo Branco Embroidery and the reflections on the clothes are very clear.

Each product tells a story. To know this technology and resort to the acquisition of the resulting products presenting the Castelo Branco embroidery and history highlights challenges for society, as they try at a social level, to involve people who are not active in society and they are given this important role in making embroidery. Still in terms of social sustainability, it foresees an increase in the supply of jobs. It is important as it deals with the cultural heritage of a region of the country, at the same time it touches on the area of sustainability.

Playponics In India—Local Hydroponics Playground Gardens Utilising Kinaesthetic Learning To Promote Global Sustainable Practices.

The paper, *Playponics in India—Local Hydroponics Playground Gardens Utilising Kinaesthetic Learning to Promote Global Sustainable Practices*, is in this chapter because it presents questions of identifiable objectives for the sustainable development for UM. The goals identifiable objectives for the sustainable development for UM are number two, End hunger, achieve food security and improved nutrition and

promote sustainable agriculture number four, ensure inclusive and equitable quality education and promote lifelong learning opportunities for all, and.

This is an important paper because this research and development project set out to investigate possible means by which future populations might have or build closer links with environmental Is-sues. The work has integrated the diverse fields of sustainability and STEM education, global products and local production. It aims to bring these aspects together in holistically considered learning environments where these elements can co-exist in engaging ways.

The paper concluded that multi-modal research is essential when we seeking to design learning systems that aim to have holistic benefits. The study has demonstrated degrees of 'validity' to the original proposition and strongly indicates high levels of in principle acceptance of the developing concept. Where the aim is to instil these understandings in children so that they may carry sustainability knowledge with them into adulthood, and where we propose integrating learning modes, mechanisms and topics as a means of achieving that, the breadth of our research needs to be as equally all-inclusive. Designers of learning systems must undertake research into each, often (on the face of it) diverse topic areas as holistically as possible, and it is this breadth of research that informs designs appropriateness for progression.

Preventing Single Use of Plastic Packaging. Design Strategies for Circular Business Models: Refill, Reuse and Recycle

Due to the changes that the circular economy approach is demanding, product design, rather than being destined for disposal, must integrate resources that should maintain their utility and value and flow back into the cycle. *Preventing single use of plastic packaging. Design strategies for circular business models: refill, reuse and recycle* shows the importance of integrating circular economy principles in an early stage of the design process. Because once the product specifications are being made, only minor changes are usually possible. Ultimately, from general design parameters, innovation in products must emerge taking into consideration human aspirations and worries—answering to a more participative, more environmental-conscient and more demanding end-user.

Design, as a discipline-oriented to study product usage and services experience become key to support business to close the loops on their business models—engaging more participative users/consumers. The acceptance that the activities of man and society have a negative impact on the planet's resources has been accepted for the first time by the user, the consumers. Nevertheless, there are huge challenges for society, citizens, business and policymakers, such as, the creation of disruptive innovation on business models for global economies, both at a manufacturing and consuming level.

Even though the content of this chapter aims to impact the sustainable development goal 8, 9 and 11, its main theme is on the 12—Responsible Consumption and

Production. Aligned not only with the Climate Action—United Nations Sustainable Development but also with the Paris Agreement. Designers, innovators, and decision-makers in businesses, aim to become active actors on the necessary shift from an industry relying on fossil resources. The study on preventing how to prevent the single use of plastic is motivated by the assumption that the anti-plastic sentiment is distracting us from the net environmental benefits of plastic and identify how design can be a source of innovation for using plastics as a highly circular resource. Accordingly, to this study, the selling drivers should prioritize services that profits from the flow of resources over time, continually reuse products and materials and using renewable energy.

Changing The Game: Social Engagement and Cultural Adaptation of Young Refugees Through Playful Design

The paper *Changing the Game: Social Engagement and Cultural Adaptation of Young Refugees through Playful Design* is aimed at understanding how design can have a positive influence on the cultural adaptation of migrant refugee children who had to flee their countries because of war, climate change and/or political or religious persecution.

Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels, this paper addresses one of the main themes of present days, the problem of forced migration, commonly referred to as seeking refuge. The target user of this study is exiled children who, in most cases, arrive at their destination alone and with few or no basic survival conditions.

The target user of this study is the exiled children who, in most cases, arrive at their destination alone and with few or no basic conditions of survival. In addition, it presents to worrying situations, as more than 69 million people worldwide have had to leave their countries of origin due to wars, climate change and/or political or religious persecution.

It is a practical example of how Design can make a difference for Society, in these case for migrant refugee children's lives, by trying to include them and educate them. These children are especially vulnerable as their basic conditions are far from optimal.

It aimed to develop an innovative solution through gaming and playing, to learning the customs and culture of the new country, preventing the users to be in dangerous situations to which they are vulnerable, also, to provide a different way of learning the new no native language. The main goal of the game is to materialize the experience of a good welcoming moment, through skills and knowledge construction.

Regarding the UN 2030 sustainable development goals this solution can promote the well-being of children, provide quality education and reduce inequalities. It is

also based on a circular economy model which promotes climate change, responsible consumption and production and sustainable growth.

Using Focus Groups to Design for Sustainable Behaviour: User-Oriented Challenges of Smart Mobility

The relevance of this article is related to the fact that "Urban mobility gained importance in the context of smart cities, considered one of its main pillars, and one aspect with a stronger impact on its sustainability." This impact depends to a large extent on the change in citizens' behaviour regarding their daily choices and decisions.

Mobility is an important requirement for social development and a core part of everyday life. Future smart mobility systems offer so much convenience and solutions for user's frustrations. However, "sharing" options which are highly important for sustainability, are not appealing to users regardless of possibilities. Plus, improving the quality of transportation detract users from walking, which is not only non-motorized, but also vital for human health. Therefore, some good intentions may also turn into undesirable. Major challenge is understanding the user's mind and changing them. If we run more design for behaviour studies like this and may explore user problems effectively, we can introduce more innovative and beneficial solutions for society.

This is the theme and contribution of the work presented in this article—which proposes to proceed with: a"step intended to identify the main obstacles that can prevent users from adopting more sustainable behaviours in what regards their urban mobility."

By contributing to change behaviours concerning the use of public/ sustainable transport options, this work can illustrate the role of Design for ODS 11—Make cities and human settlements including, safe, resilient and sustainable, namely considering the role of transport in sustainable development.

In truth, "Sustainable cities and communities" is one of the global objectives of sustainable development. By 2030, intention is to provide access to safe, affordable, accessible and sustainable transport systems. Plus, reducing the negative impact of cities, including by paying attention to air quality. This study is exploring challenges that come from user demands and frustrations related to current and future mobility modes including sustainable and unsustainable ones. Benefiting from the results, future work is motivating users to prefer and be satisfied with non-motorized mobility. It also provides data for other researchers to find possible solutions for wants and needs. In this sense, it serves the objectives for the sustainable development for UN, by revealing challenges of sustainable and smart mobility for possible solutions.

Conclusions

Design, as a project method that is transversal to all areas of knowledge and directed to the human being, leads to diversity being permanently present in everything that surrounds the designer and that is part of his projects.

On the other hand, sustainability must be appreciated according to the society for which it is designed. This means that a project completed in a defined society, according to accepted principles of sustainability, may not behave as well when moved to a different culture, that is, a different society.

As such, there is a strong need to respect basic principles that govern the environment, the economy, society and human diversity as to be able to concretize their projects, upon the request that diversity requires.

All these problems can be solved through a creative, sustainable and interdisciplinary approach. There is an increasingly call for change, coming from millions of individuals across the world.

A need for a paradigm shift towards a new economy his now requested globally. The pandemic we are living through, at this moment, puts the consumer society before a new way of looking at consumption, pointing it towards an announced end.

Design is able to question, stimulate and attempt to change the collective consciousness, in order to help us go into a new direction, a better future perhaps?

Design for society offers a systemic and holistic design ethos which integrates the frameworks and approaches of the leading design thinkers we have gathered here in these chapters.

To be able to redesign society in a way that is more intentional, equitable and inclusive, the main goal being for it to be more successful than traditional methods.

A new social and sustainable way of living, producing and consuming in a sustainable way, using concepts like circular economy and upcycling, projecting for diversity without no fear of changes of paradigms.

This is what we can read in these articles that approach not only different societies, different ways of living and different forms of seeing the results design can give to the human being, working in the direction of inclusivity and respect for the world that is the home of all life.

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